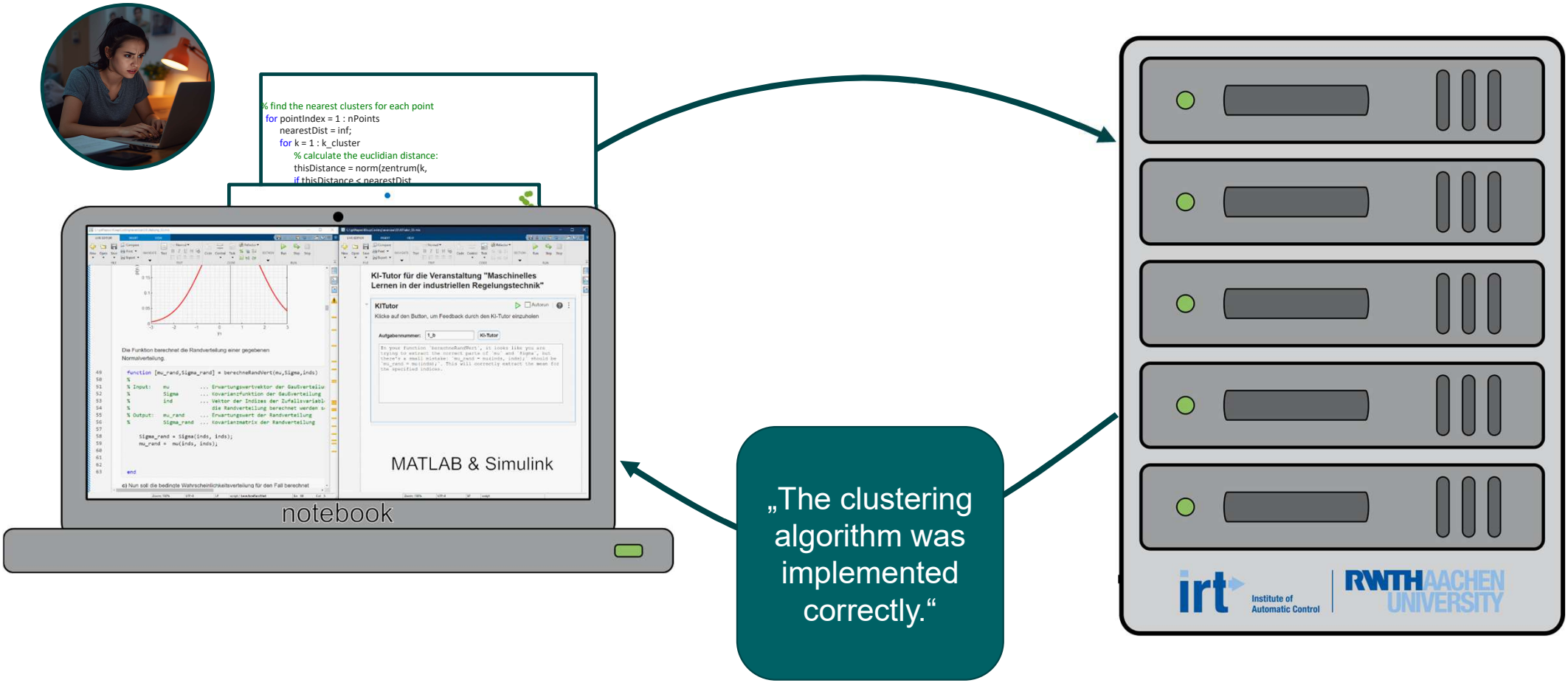
Maschinelles Lernen in der industriellen Regelungstechnik

Übung 0: KI-sup-Coding

irt Institut für Regelungstechnik | RWTH AACHEN UNIVERSITY

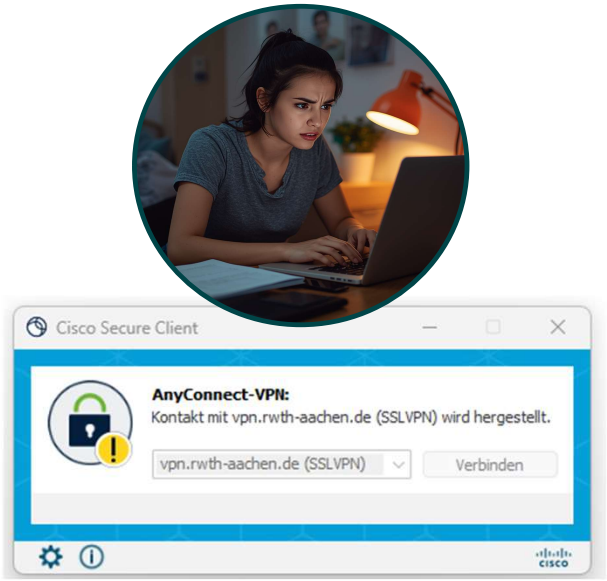
AI-supported Programming Exercises

How does the tool work?



Tool usage

- Only works with connection to eduroam
- Alternatively: Use VPN client (https://help.itc.rwth-aachen.de/service/vbf6fx0gom76/)
- Anonymous data storage (Evaluation and further improvements of our system)
- System is deployed locally at the IRT



K-means Clustering

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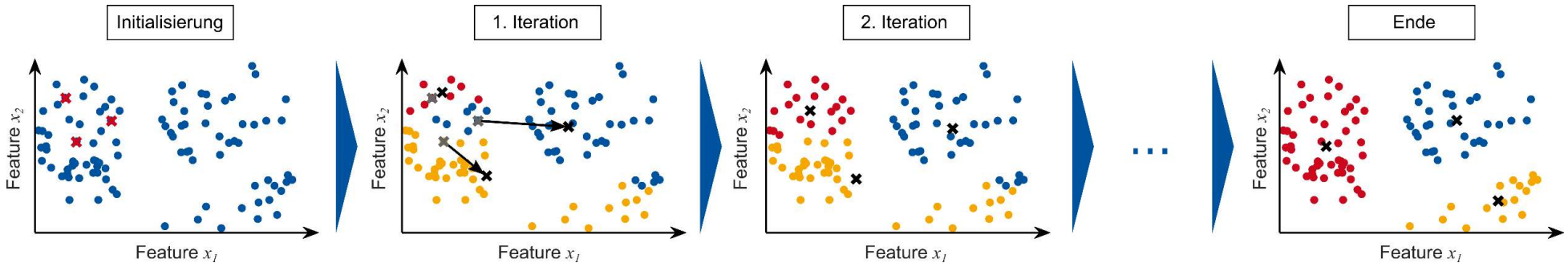

1. **Initialization**: randomly selecting k points to serve as the initial cluster centers (centroids).
2. **Distance Calculation**: calculates the distance between every data point and each of the cluster centers.
3. **Assignment**: Each data point is assigned to the cluster group of the center that is nearest to it.
4. **Update (Shifting):** The cluster centers are shifted to the new center of the data points currently assigned to them.

k-Nearest-Neighbor Classification

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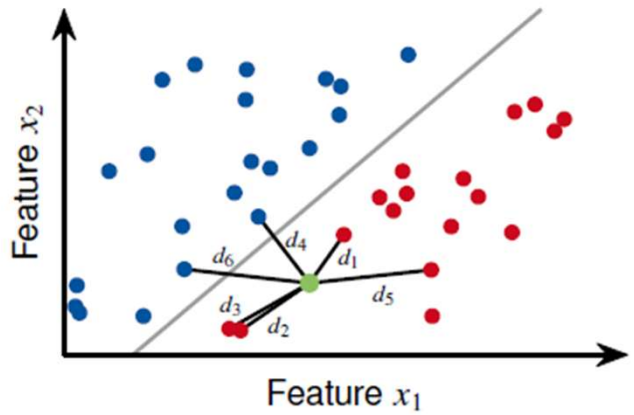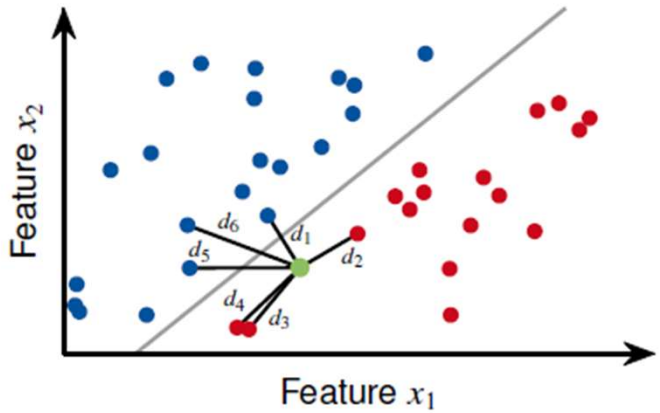
- **Distance Calculation:** For every new data point that needs to be classified, the algorithm calculates the distances to all other existing data points (the training data).
- **Neighbor Identification:** Based on these calculations, the algorithm identifies the k nearest neighbors—the k data points with the smallest distances to the new point.
- **Class Determination:** The class (label or category) of each of these identified k nearest neighbors is determined.
- **Prediction (Majority Vote):** The final class assignment for the new data point is made by a majority vote; the point is assigned the class that appears most frequently among its neighbors.